

To: The Editors-in-Chief, *Quantum Information Processing*
Re: Submission of an original research article

Dear Professors Pathak and Weinstein,

We are pleased to submit our manuscript, “**How Quantum Is the Advantage? A Fair, Calibration- and Noise-Aware Benchmark and Attribution Audit of Quantum Machine Learning for Network Intrusion Detection**,” for consideration as an original article in *Quantum Information Processing*.

The question. Quantum kernel methods are among the most widely deployed quantum machine learning algorithms, and network intrusion detection is one of the application domains in which near-perfect accuracy is most often reported for them. Whether such results reflect the quantum feature map, or the classical processing that surrounds it, is a question of quantum information science rather than of security: it asks when a quantum kernel provides an advantage that a classical kernel cannot. We answer it for a realistic learning task, with a reproducible protocol and a result we did not expect.

What we did. We benchmark hybrid variational circuits and quantum-kernel SVMs against five tuned classical baselines on four intrusion-detection datasets under one leakage-controlled protocol, with a quantum-attribution audit of parameter-matched classical controls. Two quantum results survived multiple-testing correction: a quantum kernel that out-ranked its classical surrogate under both false-discovery-rate and family-wise Holm correction, and a four-qubit hybrid that detected more attacks than a tuned Random Forest at 1% false positives. We then dissected both. The kernel result we placed on identical data with exact Gram matrices and equal tuning budgets, probed with controls on the test set and with shift induced on three further datasets, and tested with the geometric difference of Huang et al.; the hybrid result we tested against classical twins that replace only its quantum layer. Both models also ran on three IBM Heron processors.

What we found. Neither positive is quantum. The kernel advantage was produced by the classical baseline: once the surrogate’s four confounds are removed, the gap is set by the bandwidth grid searched for the classical kernel, a narrow grid giving an apparent ROC-AUC advantage of 0.167 and a wide grid 0.007, because cross-validation cannot see the shift between training and test data. Without that shift cross-validation ranks the grid correctly ($r \geq +0.94$) and the artefact disappears; holding attack families out of training on three further datasets recreates it in 26 runs, including a spurious 0.17 quantum lead, and a label-free MMD between training and unlabelled test traffic tracks when this happens. The IQP kernel lies within a geometric difference of 1.4 to 2.4 of a classical kernel on the circuit’s own phases on all four datasets. The hybrid’s trained circuit is exactly an 81-term trigonometric polynomial, and a classical tanh layer in its place reproduces its low-false-positive result on every seed; the margin over the forest comes from the input and the neural model class and lies in novel attack types. On hardware the projected kernel stays within 0.01 ROC-AUC of simulation, and the hybrid’s detection at 1% false positives is unchanged on 6,000 test rows per device and indistinguishable from its classical twin’s.

Why this journal. The contribution is a method for adjudicating claims of quantum-kernel advantage, demonstrated on a task where such claims are frequent. It builds on work that has appeared in this journal on the comparison of quantum and classical kernel functions in support vector machines (Abdulsalam and Ahmad, 2025) and on the training of quantum kernels (Innan et al., 2023), and on the broader scrutiny of quantum kernels by Schnabel and Roth, Flórez-Ablan et al. and Huang et al. Its central lessons, that statistical significance is not attribution, that a distribution shift can let the classical baseline’s tuning manufacture or erase an apparent

quantum advantage, and that a variational claim needs a classical twin of its quantum layer as its control, apply to any quantum learning benchmark.

Limitations we state up front. The variational results and the noise sweep are simulated at eight qubits; only the projected kernel and the hybrid’s quantum layer were run on hardware, as inference with 700 and 6,000 samples per device and default readout mitigation. The seed budget is small and uneven, and we report it exactly. Shift-aware selection requires attack-type labels and can only anticipate novelty of a kind the training data contains, and the MMD statistic is a coarse warning rather than a calibrated test.

Reproducibility. The code, configurations, fixed seeds, the scripts that reconstruct our exact splits, the kernel-level analysis, the raw device measurements, and the per-run result records are released at <https://github.com/Orqly-AI/quantum-ids-benchmark>; every table in the kernel analysis is generated directly from the result files. An earlier version of the benchmark, without the kernel-level analysis, is available as a preprint (arXiv:2608.18155); the kernel-level analysis is new to this submission.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission and declare no competing interests.

Thank you for your consideration.

Sincerely,

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